Numerical variability of functional MRI graph measures

Doctoral Research Proposal (ENCS 8511)

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Summary

Resting-state functional magnetic resonance imaging (rs-fMRI) and graph theory offer a non-invasive way to study network-level alterations associated with Parkinson's disease (PD). Their scientific and eventual clinical value, however, depends on results that remain reliable when reasonable analytical choices or computational environments change. As analyses become increasingly computational, ensuring their numerical reliability has become a critical challenge. Small perturbations introduced during processing can propagate through complex pipelines, leading to variability in outcomes and raising concerns about the reproducibility of reported findings. Addressing this issue requires systematic evaluation of pipeline stability to ensure results remain within acceptable numerical limits.

This doctoral research investigates the numerical variability of functional MRI graph measures. The central objective is to systematically evaluate the numerical variability of graph measures of functional connectivity and to translate that knowledge into practical guidance for robust studies.

Phase I, now completed, quantified numerical variability in graph measures derived from fMRIPrep-processed Parkinson's Progression Markers Initiative (PPMI) data. Ten Monte Carlo Arithmetic realizations were generated per participant. Numerical variability was compared with between-participant variability using the Numerical–Population Variability Ratio (NPVR). Across the evaluated local and global graph measures, mean NPVR values were approximately 0.02–0.19 and depended on graph metric, brain region, network threshold, population, and confound-regression strategy. For nominal p-values from 0.01 to 0.09, the estimated mean probability of a numerically induced significance flip was 17.7%. These results establish that computational noise is measurable and can affect inferential decisions near a significance threshold.

A second phase is currently under development, and its scope may change as the work progresses. Preliminary analyses suggest that connections close to the graph threshold and the network locations of those connections influence how small perturbations propagate into graph-measure variability. Initial experiments with a BrainNetCNN-style model indicate that the standard connectivity matrix contains predictive information for some graph measures and thresholds, while prediction remains limited for others. The proposed work will investigate whether these predictions can be improved and whether participants with comparatively high numerical sensitivity can be identified reliably.

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Chapter 1

Introduction

1.1 Parkinson's disease as a network disorder

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by both motor symptoms—including bradykinesia, rigidity, tremor, and postural instability—and non-motor symptoms involving cognition, mood, sleep, and autonomic function. Although degeneration of dopaminergic neurons in the substantia nigra is a defining pathological feature, the diversity of these manifestations cannot be explained by a single brain region. PD affects interactions among distributed cortical, subcortical, motor, cognitive, and limbic systems and is therefore increasingly understood as a network disorder [1, 2]. There is still no well-established non-invasive biomarker for early diagnosis or progression monitoring, which limits the development and evaluation of disease-modifying interventions [3].

Recent studies combining resting-state functional magnetic resonance imaging (rs-fMRI) with network neuroscience have provided valuable insights into altered brain organization in brain-related diseases. Rs-fMRI measures spontaneous fluctuations in the blood-oxygen-level-dependent signal while a participant is not performing an explicit task particularly make it more readily available and easier to share than t-fMRI as t-fMRI data varies due to the diverse task paradigms used in different studies.

Network neuroscience complements rs-fMRI by providing tools to describe how these functional connections are organized across the whole brain. Using graph theory, brain regions are represented as nodes and functional connections as edges [4, 5]. Local graph measures, including degree, clustering coefficient, betweenness centrality, and eigenvector centrality, characterize the connectivity, segregation, and influence of individual regions. Global measures, such as average shortest path length and small-worldness, characterize

large-scale integration and the balance between specialized and distributed processing. In this way, network neuroscience converts a high-dimensional rs-fMRI connectivity matrix into interpretable measures of regional and whole-network organization that can reveal distributed abnormalities missed by analyses of isolated regions.

Together, rs-fMRI and network neuroscience make it possible to investigate both where functional alterations occur and how they affect communication across the brain. Studies using this combination have reported PD-related alterations in cortico-striatal, sensorimotor, default-mode, and other intrinsic networks, as well as differences in degree, clustering, path length, efficiency, and centrality [6, 7, 2]. However, reported findings are not always consistent, and graph estimates depend on preprocessing, connectivity estimation, parcellation, thresholding, software, and computational conditions.

1.2 Motivation for reproducibility analysis

Reproducibility is essential to the accumulation of reliable scientific knowledge and is particularly important in neuroimaging, where results may ultimately inform clinical research. Open data, shared code, and detailed documentation improve transparency, but they do not guarantee that an analysis will produce consistent findings. Neuroimaging workflows contain many interconnected processing and statistical stages, so variation introduced by the data, analytical decisions, software tools, or computational environment can propagate to the final result.

One major challenge is *analytical variability*, which arises from the many defensible choices available when constructing an fMRI workflow. These choices include participant selection, acquisition and preprocessing procedures, software package and version, parameter settings, statistical models, and correction for multiple comparisons [8, 9, 10]. The Neuroimaging Analysis Replication and Prediction Study (NARPS) demonstrated the consequences of this flexibility: 70 teams analyzed the same fMRI dataset, yet no two teams used identical pipelines and, on average, 20% of teams reported different conclusions [11]. Comparisons across neuroimaging tools have similarly shown that software packages, versions, and operating systems can alter statistical maps, tissue measurements, functional components, and graph properties [12, 13].

Graph-based analyses introduce additional decisions when defining nodes and edges: parcellation and spatial resolution determine the nodes, while Pearson correlation, partial correlation, mutual information, and other connectivity estimators can produce different network architectures [5, 14]. Reproducibility also depends on preprocessing choices such as

global signal regression and smoothing, as well as graph sparsity or density; previous reviews have therefore reported reliability ranging from poor to moderate for some rs-fMRI graph measures [15].

A second, less visible challenge is *numerical variability*. Small differences in floating-point calculations can be introduced by hardware architectures, operating systems, compilers, mathematical libraries, or parallel execution. Although these perturbations begin at very small numerical scales, they can accumulate through long or nonlinear pipelines and produce meaningful differences in derived outcomes [13, 16, 17, 18]. Numerical variability may therefore contribute to the broader analytical variability observed across tools and infrastructures [19].

1.3 Research problem

Network neuroscience provides a powerful framework for studying the mechanisms underlying brain-related diseases. As analyses become increasingly computational, ensuring their numerical reliability has become a critical challenge.

While the numerical variability of structural imaging workflows has been investigated, with findings ranging from negligible to substantial, functional MRI (fMRI) pipelines and their derived graph measures remain underexplored. Without rigorous stability assessment, conclusions drawn from these measures may remain uncertain.

This research addresses the gap by evaluating the numerical variability of graph measures of fMRI derived from the widely used fMRIPrep pipeline. It investigates how numerical perturbations influence the reliability of fMRI derived graph metrics, providing insights about the robustness of network-based biomarkers and contributes to best practices for computational reproducibility in neuroimaging research.

The first major contribution of this thesis is systematic evaluation of the numerical variability of graph measures of functional connectivity derived from the widely used fMRIPrep pipeline and compared it to population variability. The resulting Numerical-Population Variability Ratio (NPVR) values typically ranged from 0.02 to 0.19 for most graph metrics, indicating a measurable influence of numerical variability on network-derived outcomes. NPVR values varied across brain regions, thresholding choices, and confound regression strategies. For results close to the significance threshold, propagation of the observed NPVR values indicated an average 17.7% probability that numerical variability would change a result from significant to non-significant, or vice versa. These findings highlight numerical variability as an important factor in functional network studies, particularly when examining

subtle effects or working with small sample sizes.

This thesis proposal is organized around its main contributions, with each chapter addressing a distinct aspect of numerical uncertainty in numerical variability of graph measures of fMRI neuroimaging analyses. Before describing these contributions in detail, a theoretical background on numerical variability is provided in Chapter 2. Chapter 3 describes the methods used to evaluate numerical variability in fMRI graph measures, including the datasets, preprocessing pipelines, and graph metrics analyzed. Chapter 4 presents the proposed future work and subsequent research phases.

Chapter 2

Background

2.1 Numerical variability

As neuroscience has transitioned into an increasingly computational field, it has encountered challenges related to numerical reproducibility similar to those observed in many other scientific domains [17]. In neuroimaging, where analyses must be re-executed to reproduce findings and evaluate their validity, numerous factors can contribute to variability between results. Data noise, software and hardware updates, operating-system differences, mathematical libraries, and parallelization strategies can introduce small numerical perturbations that produce deviations in the results and potentially compromise the reliability of computational analyses [13, 16]. Moreover, operations commonly employed in neuroimaging, including registration, segmentation, interpolation, and brain-network modelling, can be unstable when subjected to minor perturbations in the data or analytical pipeline [17]. Numerical instability is therefore a fundamental consideration in computational reproducibility, and evaluating the resulting numerical variability is important for establishing the reliability of neuroimaging results.

2.1.1 Sources of numerical variability

Numerical errors may occur when non-integer data are represented using the IEEE 754 floating-point standard, which is widely employed to represent real numbers and perform arithmetic operations in computer systems. A floating-point number consists of a sign, an exponent, and a fraction. In single precision (32-bit), one bit is allocated to the sign, eight bits to the exponent, and 23 bits to the fraction; in double precision (64-bit), the corresponding allocations are one, 11, and 52 bits. Although this representation supports a

broad range of very small and very large values, many real numbers have non-terminating binary representations and must therefore be approximated within the available number of bits [16].

Two common consequences of finite-precision arithmetic are rounding error and catastrophic cancellation. Rounding error occurs when the result of an operation is rounded to fit within the available precision, producing a small deviation from the exact mathematical value. Catastrophic cancellation occurs when two nearly equal quantities are subtracted, causing a substantial loss of significant digits. These errors may accumulate across successive operations and can affect the accuracy and stability of a computation, particularly in long or nonlinear pipelines. Consequently, theoretically equivalent implementations may produce different results because of differences in operation order, compiler optimization, hardware architecture, or execution environment [13, 17].

2.1.2 Assessment of numerical variability

Stochastic arithmetic, uncertainty or sensitivity analysis, and random-seed analysis are among the methods available for detecting numerical instability. This thesis employs stochastic arithmetic, an approach used to investigate the effects of finite-precision arithmetic by modelling and estimating losses of numerical accuracy. A widely used technique within stochastic arithmetic is Monte Carlo Arithmetic (MCA), which models floating-point error through controlled random perturbations [20, 16]. For a floating-point value x, MCA defines an inexact counterpart as

$$\text{inexact}(x, t) = x + 2^{e_x - t}\xi, \tag{2.1}$$

where e_x is the exponent of x, t is the virtual precision, and ξ is a uniformly distributed random variable on $(-1/2, 1/2)$. MCA can be applied in three perturbation modes. Precision Bounding perturbs the inputs of an operation and enables the detection of catastrophic cancellation; Random Rounding perturbs the output and enables the investigation of rounding error; and Full MCA combines both modes [16].

Under the MCA framework, floating-point operations in an algorithm or software pipeline are randomly perturbed, and the analysis is repeated to generate a distribution of equally plausible results. The variability of this distribution provides information about the numerical stability of the evaluated software or pipeline. Numerical uncertainty may be summarized through the dispersion of repeated results, the number of significant digits that remain consistent across executions, or outcome-specific measures that compare numerical variability

with other relevant sources of variation. The resulting estimates depend on the perturbation mode, the virtual precision, and the extent of software instrumentation [21, 22].

2.1.3 Evaluation of MCA

After obtaining samples through stochastic arithmetic, we quantify the numerical uncertainty of the results using both the number of significant bits and the standard deviation across MCA realizations.

Significant Bits:

In the context of MCA, the number of significant bits can be interpreted as the number of bits expected to agree across repeated executions of the same computation under stochastic arithmetic [23].

Let $X_1, X_2, \ldots, X_n$ denote the values obtained from n independent executions of the program using stochastic arithmetic. These values are considered realizations of the same random variable X. The expected value $\mu = E[X]$ can be estimated using the empirical mean

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i,$$

and the variability of the realizations can be estimated using the empirical standard deviation

$$\hat{\sigma} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i - \hat{\mu})^2}.$$

The MCA number of significant bits, interpreted as a measure of numerical precision, is estimated as [23]

$$s_{\widehat{\mathrm{MCA}}} = -\log_2 \left| \frac{\hat{\sigma}}{\hat{\mu}} \right|,$$

where $s_{\widehat{\mathrm{MCA}}}$ approximates the number of significant bits from the n MCA realizations, and $\hat{\mu}$ and $\hat{\sigma}$ denote their empirical mean and standard deviation, respectively.

In addition to the number of significant bits, we directly quantify numerical variability using the standard deviation of each output metric across MCA realizations. For a given metric, a larger $\hat{\sigma}$ indicates greater variation induced by numerical perturbations. Thus, while $s_{\widehat{\mathrm{MCA}}}$ provides a relative measure of numerical precision by normalizing the variability

by the magnitude of the estimated value, $\hat{\sigma}$ provides an absolute measure of the numerical variability of the metric across MCA executions.

2.2 Numerical variability in neuroimaging: literature review

Early evidence of numerical variability in neuroimaging arose from direct comparisons of computational environments. Glatard et al. [13] evaluated FSL, FreeSurfer, and CIVET across different software builds and GNU/Linux operating systems. Although many outputs were similar, substantial differences occurred in particular analyses: Dice coefficients for FSL tissue classifications reached values as low as 0.59, independent-component analyses of fMRI data differed across operating systems for approximately one-third of the evaluated participants, and regional cortical-thickness estimates varied with the software build or operating system. The fMRI differences were traced to variations in motion-correction outputs. This study established that computational infrastructure can affect scientifically relevant neuroimaging results and emphasized the need to evaluate the numerical stability of complete pipelines.

Subsequent work used controlled perturbations to characterize this variability rather than relying exclusively on comparisons between existing environments. Kiar et al. [22] instrumented deterministic and probabilistic structural-connectome pipelines with MCA and evaluated the resulting connectomes and network features. Stability ranged from nearly complete agreement to only zero or one significant digit. Although some aggregate properties of network organization remained stable, individual edge weights and participant-level features could be unreliable, demonstrating that group-level robustness does not necessarily imply reliability at the individual level. Salari et al. [16] subsequently introduced fuzzy libmath to perturb calls to mathematical libraries and reproduce variability caused by operating-system updates in Human Connectome Project preprocessing pipelines. The simulated and observed operating-system variability were comparable, while between-participant differences were preserved. However, the processed images retained fewer than eight significant bits of the 24 available, providing further evidence that infrastructure-level perturbations can propagate through neuroimaging workflows.

Numerical variability has also been investigated as a source of potentially useful variation. Using MCA-perturbed structural connectomes, Kiar et al. [24] constructed augmented datasets for age classification. Resampling across plausible perturbed networks improved test performance and generalizability across the evaluated classifiers, preprocessing methods,

and dimensionality-reduction techniques. The improvement did not require a large number of perturbations. These results showed that numerical variability is not only a threat to reproducibility; when it is explicitly characterized, it can also be used to reduce computational bias and improve the robustness of downstream models.

Another line of work has incorporated numerical variability into software validation. Chatelain et al. [18] defined acceptable bounds of result variation from Random Rounding samples and used these bounds to construct stability tests for fMRIPrep. The tests accepted variations produced by a reference software version while detecting subtle image-processing changes and corrupted templates. This approach extends conventional software testing by recognizing that exact bitwise equality is not always an appropriate criterion for scientific pipelines. It also provides a mechanism for evaluating whether results obtained after software or hardware changes remain within the expected numerical variability of the reference analysis.

Recent studies have extended numerical-stability analysis to deep-learning-based neuroimaging. Gonzalez-Pepe et al. [25] applied Random Rounding during inference with SynthMorph and FastSurfer and compared the results with the conventional FreeSurfer pipeline. Across 32 participants, CNN inference was more numerically stable: nonlinear-registration outputs retained approximately 19 significant bits compared with 13 for the reference pipeline, and whole-brain segmentations achieved mean Dice scores of 0.99 compared with 0.92. In contrast, their later investigation of FastSurfer training found substantial training-time variability, particularly in cortical regions, which exceeded that of FreeSurfer [26]. Random-seed variability exhibited a spatial structure similar to numerical variability and could be used to construct ensembles that improved brain-age regression. Together, these findings distinguish the comparatively stable inference stage from the more variable iterative training stage.

Numerical variability can also depend strongly on the algorithm and its configuration. Mirhakimi et al. [27] compared linear registration with SPM, FSL, and ANTs across similarity measures, templates, and cohorts of healthy participants and participants with PD. SPM was the most stable under its default configuration, whereas FSL and ANTs exhibited greater variability, and ANTs occasionally produced registration failures under perturbation. Stability did not differ significantly between the healthy and PD cohorts, suggesting that evaluations conducted in healthy populations may generalize to clinical samples for this task. The study also demonstrated that numerical-variability measures may support automated quality control, connecting computational stability with the detection of problematic registrations.

Most recently, Chatelain et al. [28] quantified the practical impact of numerical variability on structural MRI measures in PD using the Numerical–Population Variability Ratio. For

several cortical and subcortical outcomes, numerical variability approached one-third of between-participant variability and altered conclusions concerning group differences and clinical associations. Propagation of numerical variability to 13 previously published PD studies produced an estimated mean probability of a significance-classification change of approximately 5% for cross-sectional analyses and 10% for longitudinal analyses. This work moves beyond reporting numerical differences by relating them directly to population variability and inferential decisions.

Taken together, these studies demonstrate that numerical variability can arise across operating systems, mathematical libraries, conventional image-processing algorithms, connectome pipelines, and deep-learning models. Its magnitude is specific to the pipeline, outcome, participant, and analytical configuration, and stability at the group level does not guarantee stability of regional or participant-level measures. Nevertheless, most previous investigations have focused on structural or diffusion MRI, image registration, segmentation, or structural connectivity. The numerical variability of functional-connectivity graph measures remains comparatively underexplored, motivating the Phase I study presented in Chapter 3.

Chapter 3

Numerical variability of functional MRI graph measures

Abstract

Network neuroscience provides a powerful framework for studying the mechanisms underlying brain-related diseases. As analyses become increasingly computational, ensuring their numerical reliability has become a critical challenge. Small perturbations introduced during processing can propagate through complex pipelines, leading to variability in outcomes and raising concerns about the reproducibility of reported findings. Addressing this issue requires systematic evaluation of pipeline stability to ensure results remain within acceptable numerical limits. While the numerical variability of structural imaging workflows has been investigated, with findings ranging from negligible to substantial, functional MRI (fMRI) pipelines and their derived graph measures remain underexplored. Without rigorous stability assessment, conclusions drawn from these measures may remain uncertain. We systematically evaluated the numerical variability of graph measures of functional connectivity derived from the widely used fMRIPrep pipeline and compared it to population variability. The resulting Numerical-Population Variability Ratio (NPVR) values typically ranged from 0.02 to 0.19 for most graph metrics, indicating a measurable influence of numerical variability on network-derived outcomes. NPVR values varied across brain regions, thresholding choices, and confound regression strategies. For results close to the significance threshold, propagation of the observed NPVR values indicated an average 17.7% probability that numerical variability would change a result from significant to non-significant, or vice versa. These findings highlight numerical variability as an important factor in functional network studies, particularly when examining subtle effects or working with small sample sizes.

Materials and methods

We applied the widely used fMRIPrep preprocessing pipeline to fMRI data from the PPMI dataset to assess numerical variability, focusing on cross-sectional analyses. Numerical variability was evaluated using Monte Carlo Arithmetic (MCA) [20, 21], a technique that characterizes the sensitivity of a computational pipeline to small numerical perturbations. By repeatedly perturbing floating-point operations, MCA enables us to quantify how much variability arises purely from numerical errors. We then examined how these perturbations propagate into derived graph measures and compared the resulting numerical variability to inter-individual variability, evaluating its potential impact on downstream statistical analyses.

Ethics statement

Concordia University's Office of Research confirmed that this project was exempt from institutional research ethics approval. The analyses were conducted in accordance with the PPMI Data Use Agreement. The PPMI data were originally collected with approval from the relevant institutional review boards at participating sites, and all participants provided written informed consent.

Dataset

We used data from the *Parkinson Progression Marker Initiative* (PPMI) [3], an ongoing, international, multicenter observational study designed to identify biomarkers of Parkinson's disease (PD). From the PPMI cohort, we selected individuals with available resting-state fMRI (rs-fMRI) data, including both Parkinson's disease (PD) patients and healthy control (HC) participants. The PPMI dataset used in this study was acquired across multiple imaging sites using 3T MRI scanners from different manufacturers, including Siemens, GE Healthcare, and Philips. The scanner manufacturers and models represented in the resting-state fMRI and structural MRI datasets are summarized in Table A in S1 File. PPMI provides multiple rs-fMRI acquisitions using 2D gradient-echo T2*-weighted EPI with different acquisition configurations (e.g., `epi_2d`, RL, PA) and phase-encoding directions. To avoid additional variability introduced by mixing acquisition types and phase-encoding directions, we restricted our analysis to the RL runs only, including 38 of healthy controls and 147 of PD patients. The RL rs-fMRI acquisition consists of 240 volumes with 10 min total scan time, with a repetition time (TR) of 2500 ms, an echo time (TE) of 30 ms, a slice thickness of 3.5 mm, 40 axial slices, a field of view (FOV) of 224×224 mm, and a matrix size of 64×64. Participants

were instructed to keep their eyes open and remain still during the scan. Since the data were in DICOM format, they were converted to NIfTI format using the HeuDiConv conversion framework (version 1.2) to enable compatibility with the preprocessing pipeline, which requires BIDS-validated input data.

The data were accessed multiple times between August and September 2024, and we did not receive access to identifiable information.

Data processing

All anatomical and functional data were preprocessed using *fMRIPrep* (version 23.2.1). Anatomical T1-weighted images underwent intensity non-uniformity correction, skull stripping, tissue segmentation, and nonlinear normalization to the MNI152NLin2009cAsym template.

Functional preprocessing included the generation of a BOLD reference volume, head-motion correction, and co-registration of the BOLD reference to the T1-weighted image using boundary-based registration. The corrected BOLD time series were subsequently transformed into MNI space using the composed anatomical and functional transformations. fMRIPrep also produced a comprehensive set of confound regressors, including motion parameters, framewise displacement (FD), DVARS, global signals, and components derived from aCompCor and tCompCor. All spatial resampling steps were combined into a single interpolation using the composed transforms.

Following preprocessing, functional connectomes were generated using `Nilearn`. Regional time series were extracted using a `NiftiLabelsMasker` applied to the preprocessed BOLD images, using the Schaefer 2018 parcellation with 100 cortical regions and 7 functional networks (`fetch_atlas_schaefer_2018`, $n_\text{ROIs} = 100$, $n_\text{networks} = 7$). Spatial smoothing (6 mm FWHM) and temporal standardization were applied during masking. Because the optimal strategy for confound regression in resting-state fMRI remains an active area of investigation, with different denoising approaches involving distinct trade-offs and differentially affecting estimates of functional connectivity [29, 30, 31, 32], we evaluated both preprocessing variants to determine whether numerical variability depends on this consequential preprocessing choice.

Two versions of the functional connectome were therefore constructed for each subject and each MCA realization:

1. **With confound regression**: time series were extracted after regressing out the six rigid-body motion parameters (three translations and three rotations) provided by fMRIPrep.

2. **Without confound regression**: time series were extracted without applying any confound regression, isolating numerical variability arising solely from preprocessing.

For each version, Pearson correlation matrices were computed using Nilearn's `ConnectivityMeasure` (kind=`"correlation"`), yielding one functional connectivity matrix per subject per MCA run.

Graph construction and graph-theoretical measures

After generating functional connectomes, the resulting matrices were thresholded based on absolute correlation values, as commonly performed in resting-state functional connectivity studies, and converted into binary, undirected adjacency matrices.

Specifically, absolute correlation thresholds of 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5 were applied. An edge was assigned between regions i and j whenever

$$|r_{ij}| >= T,$$

where T denotes the selected correlation threshold. Otherwise, no edge was assigned. The selected thresholds span a broad range of network densities. Lower thresholds retain a larger number of weak functional connections and therefore produce denser graphs, whereas higher thresholds retain only stronger connections and produce progressively sparser graphs. Because graph-theoretical measures are known to depend on network density and no universally accepted correlation threshold exists for resting-state functional connectivity analysis, multiple thresholds were evaluated to characterize the threshold dependence of numerical variability.

From these networks, we computed four local graph metrics: degree centrality, clustering coefficient, betweenness centrality, and eigenvector centrality, as well as two global metrics: small-worldness and average shortest path length. All graph measures were computed using `Python` (version 3.12) and `NetworkX` (version 3.5) library.

Degree centrality quantifies the direct connectivity of a node by counting its number of adjacent neighbors, reflecting local influence.

Eigenvector centrality extends this notion by assigning higher importance to nodes connected to other highly connected nodes, capturing their global prominence in the network [2].

Betweenness centrality measures the fraction of shortest paths that pass through a node, indexing its role in mediating information flow.

Clustering coefficient reflects functional segregation by quantifying how strongly a

node's neighbors are interconnected [7].

For global topology, we computed the **average shortest path length**, which measures the efficiency of information integration across the entire network, and **small-worldness**, defined using normalized clustering and path-length relative to matched random networks, capturing the balance between segregation and integration characteristic of small-world architecture [6]. Together these metrics summarize a variety of information transmission qualities of the network.

Evaluation of numerical variability

Fuzzy libmath is a lightweight implementation that restricts MCA perturbations to the outputs of elementary mathematical library functions, such as `exp`, `log`, `sin`, and `cos`, whose implementations may vary across mathematical libraries, operating systems, hardware architectures, and execution settings [13]. Fuzzy libmath was loaded using the `LD_PRELOAD` mechanism, which intercepts calls to the standard mathematical library and redirects them to the instrumented implementation. Targeting these functions therefore provides a practical model of machine-level numerical variability while limiting instrumentation overhead [16]. The virtual precision was set to $t = 24$ bits for binary32 values and $t = 53$ bits for binary64 values. Thus, perturbations were introduced at the least significant bit of the corresponding native floating-point representation.

By repeatedly executing the *fMRIPrep* pipeline using fuzzy libmath (10 MCA realizations per subject), we obtained a distribution of perturbed outputs representing equally plausible numerical outcomes arising from finite-precision arithmetic.

Across the 10 repetitions, the input images, software versions, and processing parameters were held constant, while each execution used an independent realization of the stochastic perturbations. Executing the complete preprocessing pipeline for 10 independent MCA realizations therefore resulted in an overall computational workload approximately ten times that of a single execution. However, because the MCA realizations are independent, they can be executed in an embarrassingly parallel fashion. Thus, when sufficient computational resources are available, the realizations can be run concurrently, substantially reducing the wall-clock time associated with the repeated executions. On these datasets, MCA to IEEE wall-clock processing time was 1.29. Thus, a single MCA execution required approximately 29% more wall-clock time than the corresponding IEEE execution. The aggregate processing requirement of the 10 MCA repetitions was approximately 12.9 times that of one IEEE execution.

Numerical variability measures

To evaluate the impact of numerical variability on graph measures, we used the Numerical–Population Variability Ratio (NPVR or ν_{npv}), a framework developed in [28]. This framework quantifies numerical variability relative to variability across subjects, providing a formal comparison between the variability induced by MCA perturbations and the variability observed across subjects.

The $\nu_{\text{npv}}$is defined as:

$$\nu_{\text{npv}} = \frac{\sigma_{\text{num}}}{\sigma_{\text{pop}}}$$

where σ_{num} represents numerical variability across MCA repetitions for each individual subject and σ_{pop} represents inter-subject differences within each repetition. For each region of interest, $x_{i,j}$ represents the measurement for subject j in MCA repetition i. Numerical variability quantifies intra-subject measurement consistency:

$$\sigma^2_{\text{num}} = \frac{1}{m}\sum_{j=1}^{m}\left[\frac{1}{n-1}\sum_{i=1}^{n}(x_{i,j} - \bar{x}_{\cdot,j})^2\right] \tag{3.1}$$

Population variability captures inter-subject differences:

$$\sigma^2_{\text{pop}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{m-1}\sum_{j=1}^{m}(x_{i,j} - \bar{x}_{i,\cdot})^2\right] \tag{3.2}$$

where $\bar{x}_{\cdot,j}$ and $\bar{x}_{i,\cdot}$ denote column and row means, n is the total number of MCA repetitions and m is the number of subjects. Higher ν_{npv}values indicate regions where computational variability potentially compromises the detection of true population differences. ν_{npv} can also be propagated to statistics used in common statistical tests. For instance, the standard deviation of Cohen's d (σ_d) can be approximated as follows [28]:

$$\sigma_d = \frac{2}{\sqrt{n}}\,\nu_{\text{npv}}, \tag{3.3}$$

This relation formalizes how numerical variability impacts the reliability of downstream inferences. The numerical variability of other classical statistics, including the t-statistic and the F-statistic, as well as of their associated p-values, can also be expressed as a function of ν_{npv}.

Propagation of NPVR to downstream statistical inference

To translate the observed NPVR values into measures of statistical-inference variability, we propagated the NPVR associated with each graph metric and network threshold to statistics commonly used in neuroimaging. These included Cohen's d, the two-sample t-test, partial correlation, and ANCOVA. The first-order propagation formulas developed in [28] were used to estimate the numerical standard deviation of each statistic and, where applicable, the numerical uncertainty of its associated p-value, denoted by σ_p. The formulas are summarized in Table B in S1 File, and their complete mathematical derivations and numerical validation are provided in [28].

We used sample sizes $n = 30, 50, 75, 100, 300, 500$, and 1000, together with representative test-statistic values, to compute the nominal p-values, p_0, and their propagated numerical uncertainties, σ_p, for two-sample t-tests, partial correlations, and ANCOVA. The representative values were $t = 0.5, 1, 2, 3, 4, 5$, and 6 for two-sample t-tests; $r = 0.05, 0.1, 0.2, 0.3, 0.4$, and 0.5 for partial correlations; and $F = 0.5, 1, 2, 4, 6, 8, 10$, and 15 for ANCOVA. For each combination of NPVR, sample size, and representative test statistic, p_0 was computed from the corresponding statistical distribution, while σ_p was obtained by propagating the NPVR associated with each graph metric and network threshold using the formulas in Table B in S1 File. These representative values were selected to cover a range of weak to strong statistical effects and are intended as practical reference values rather than universal thresholds. For the two-sample t-test and partial-correlation calculations, the degrees of freedom were set to $\mathrm{df} = n$. For ANCOVA, they were set to $\mathrm{df}_1 = 1$ and $\mathrm{df}_2 = n$.

Following the approach in [28], the numerically perturbed p-value was modeled as a random variable $P \sim \mathrm{Beta}(a, b)$. The Beta parameters were determined by moment matching such that $\mathrm{E}[P] = p_0$ and $\mathrm{Var}(P) = \sigma_p^2$. Statistical significance was evaluated at $\alpha = 0.05$. The probability that numerical variability changes the significance classification was calculated as

$$P_{\mathrm{flip}} = \begin{cases} 1 - F_{\mathrm{Beta}}(\alpha; a, b), & p_0 \leq \alpha, \\ F_{\mathrm{Beta}}(\alpha; a, b), & p_0 > \alpha, \end{cases} \tag{3.4}$$

where $F_{\mathrm{Beta}}(\alpha; a, b)$ is the cumulative distribution function of the fitted Beta distribution evaluated at α.

For a nominally significant result, P_{flip} represents the probability that numerical variability shifts the p-value above the significance threshold, indicating false-positive risk. For a nominally non-significant result, it represents the probability that numerical variability shifts the p-value to or below the threshold, indicating false-negative risk. These labels

describe instability of the nominal significance classification rather than classical Type I or Type II error rates because the underlying ground truth is unknown. The distance from the significance threshold was expressed as $p_0 - \alpha$.

Results

The influence of numerical variability on statistical inference

To illustrate how numerical noise can influence statistical inference and experimental design, we generated a series of simulations (Fig. 3.1). As shown in Figs. 3.1A–B, two synthetic populations of ten subjects each were generated under "low" and "high" within-individual variability conditions with identical underlying subject-level means. These scenarios emulate differences in the numerical variability of derived measurements and allow us to quantify their effects on inference.

The resulting Numerical–Population Variability Ratio (NPVR; Fig. 3.1C) quantifies the magnitude of numerical variability relative to inter-subject variability, thereby providing a direct measure of the numerical stability of the derived measurements.

Fig. 3.1D further illustrates how numerical variability propagates into uncertainty in Cohen's d. According to Eq. 3.3, σ_d increases linearly with NPVR at a fixed sample size and decreases proportionally to $1/\sqrt{n}$ as sample size increases. Consequently, analyses with higher NPVR values exhibit greater numerical uncertainty in estimated effect sizes and require larger sample sizes to attain the same level of numerical precision as analyses with lower NPVR values.

Numerical variability varies across graph statistics and network thresholding decisions

Figs. 3.2 and 3.3 present the NPVR of local and global graph metrics across network thresholds. For each metric, each plot shows the mean NPVR for the pooled PD+HC sample and separately for the PD and HC groups. For local metrics, NPVR was averaged across the 100 nodes, whereas the global metrics yield one value per network. Numerical and inter-subject variability values are additionally shown for the pooled PD+HC sample.

Magnitude of NPVR: For local graph metrics (Fig. 3.2), the average NPVR ranged approximately from 0.04 to 0.19, while for global metrics (Fig. 3.3), it ranged approximately from 0.02 to 0.18. NPVR therefore remained below 0.2 across all investigated graph metrics

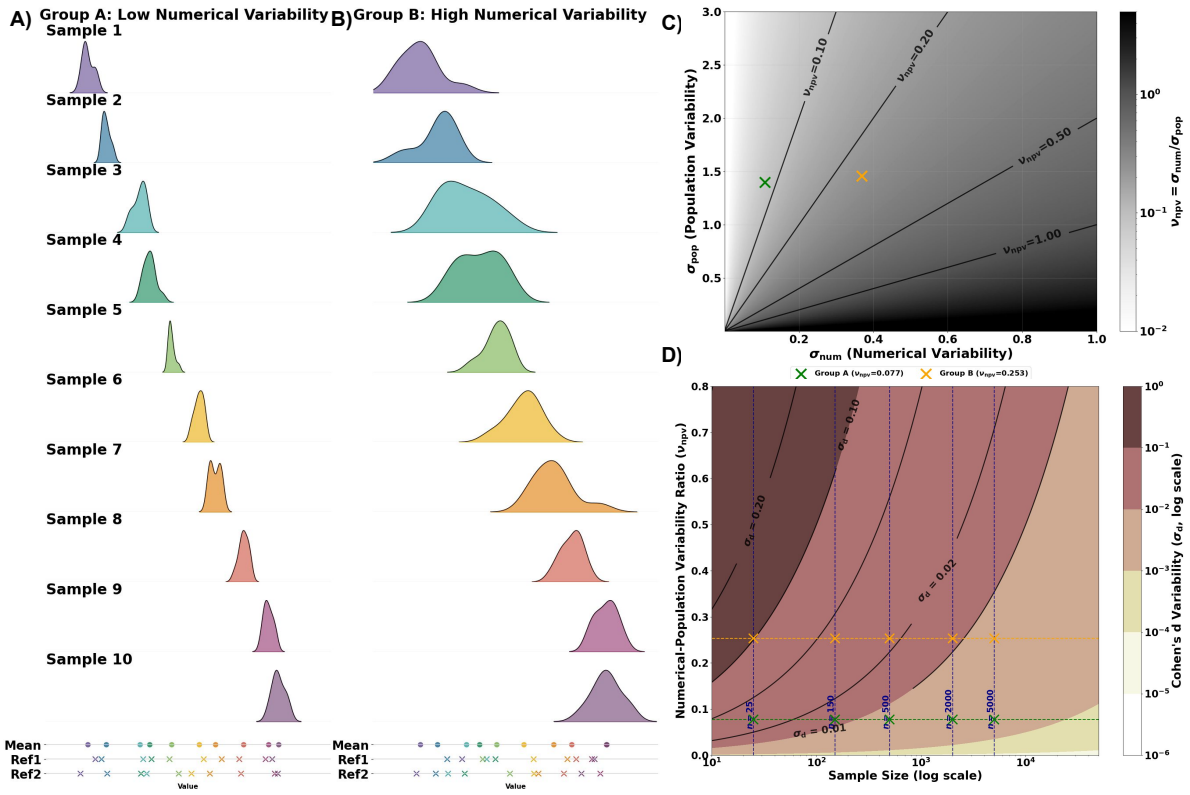


Figure 3.1: **Impact of numerical variability on inference. (A–B)** Simulated distributions of measurement values from two populations (10 subjects each) under low (A) and high (B) within-individual numerical variability. **(C)** Numerical-to-population variability ratio (NPVR; ν_{npv}) computed for the two populations. **(D)** Propagated numerical standard deviation of Cohen's d, $\sigma_d = 2\nu_{\text{npv}}/\sqrt{n}$, as a function of sample size and NPVR. Darker shading indicates greater numerical uncertainty in Cohen's d, and lighter shading indicates lower uncertainty. Vertical dashed lines indicate selected example sample sizes.

and thresholds, although its magnitude and threshold dependence differed substantially across metrics. For local metrics, the NPVR increased with threshold for *clustering coefficient* and *betweenness centrality*, suggesting that numerical variability grew faster than inter-subject variability. Conversely, for *degree* and *eigenvector centrality*, the NPVR decreased, indicating that inter-subject variability dominated. For global metrics, *small-worldness* and *average shortest path length* showed higher NPVR at a threshold of 0.05, followed by a decrease up to 0.3, and a subsequent rise—more pronounced for the latter for the pooled sample. The PD- and HC-specific NPVR values are also shown directly in Figs. 3.2 and 3.3 (red and blue stars, respectively), alongside the pooled PD+HC values (black stars). Although differences in NPVR magnitude were observed between PD and HC for several graph metrics, the main threshold-dependent patterns were broadly preserved across the two populations. For the local metrics, HC generally exhibited lower NPVR than PD, particularly at higher

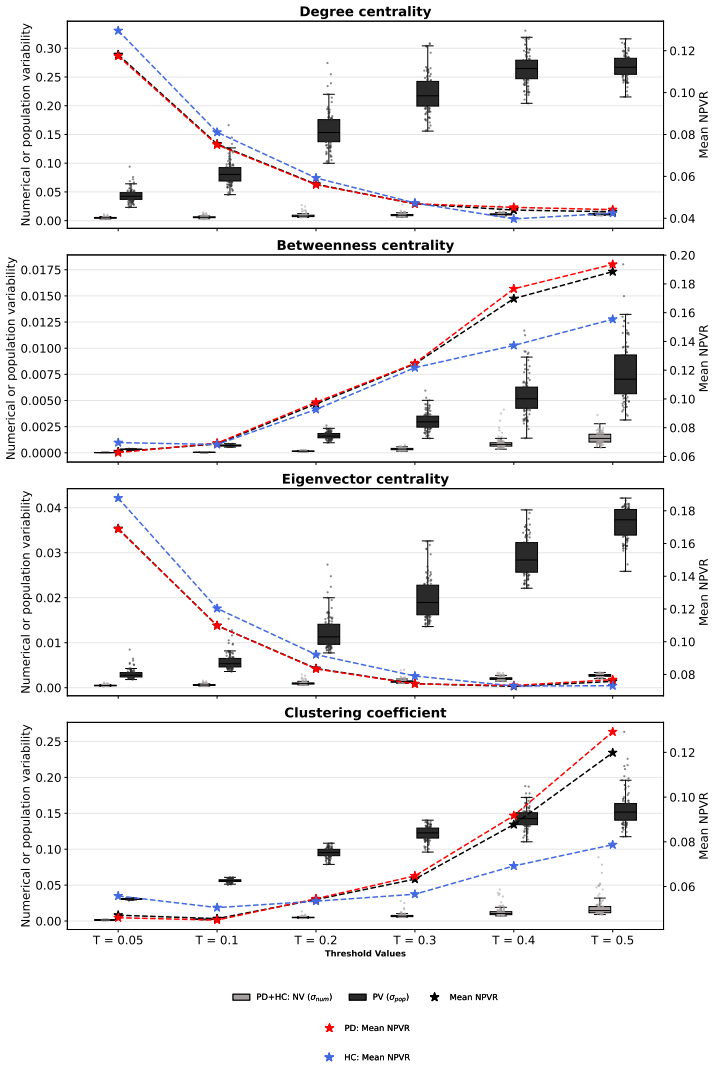


Figure 3.2: **Numerical-to-population variability ratio for local graph metrics.** Numerical variability (NV; σ_{num}), population variability (PV; σ_{pop}), and the mean numerical-to-population variability ratio (NPVR; $\bar{\nu}_{\text{npv}}$) across network thresholds. NPVR is shown for the pooled Parkinson's disease (PD) and healthy control (HC) sample and separately for the PD and HC groups. Each point in a boxplot represents one brain region.

thresholds for betweenness centrality and clustering coefficient. Betweenness centrality and clustering coefficient showed a similar increasing pattern with threshold; degree and eigenvector centrality showed a similar decreasing pattern with threshold in both groups. For the global metrics, the group differences were relatively small for small-worldness, whereas average shortest path length showed more pronounced differences between PD and HC at some thresholds, particularly at $T = 0.4$ and $T = 0.05$. Because the pooled sample contained substantially more PD than HC participants (147 PD versus 38 HC), the pooled NPVR estimates were generally closer to the PD estimates than to the HC estimates.

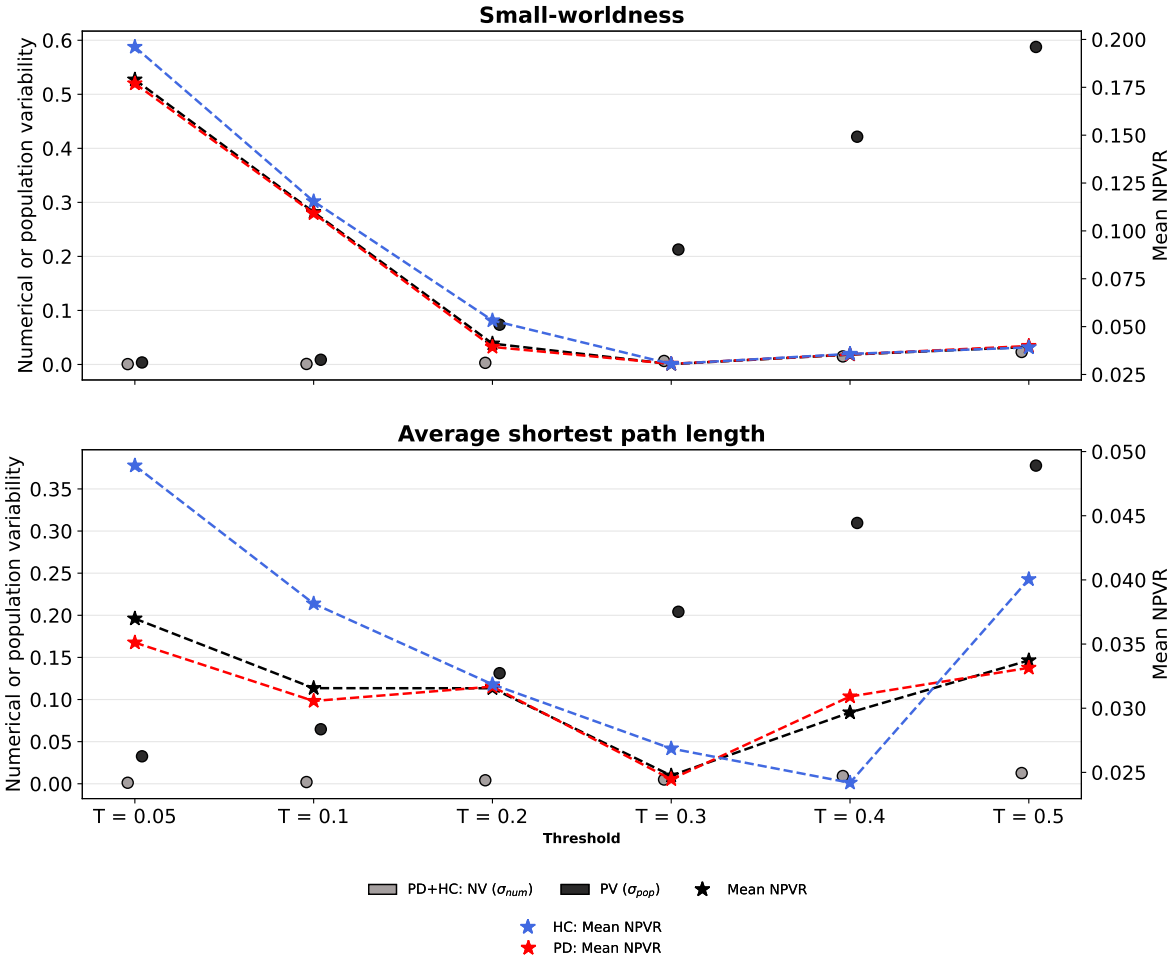


Figure 3.3: **Numerical-to-population variability ratio for global graph metrics.** Numerical variability (NV; σ_{num}), population variability (PV; σ_{pop}), and the mean numerical-to-population variability ratio (NPVR; ν^{-}_{npv}) across network thresholds. NPVR is shown for the pooled Parkinson's disease (PD) and healthy control (HC) sample and separately for the PD and HC groups.

Regional variation in normalized regional NPVR maps across metrics and thresholds interacts with threshold decisions

Fig. 3.4 shows regional maps of the NPVR (numerical-to-inter-subject variability) for the four local graph metrics across thresholds. Overall, the NPVR is spatially heterogeneous and depends both on the metric and on the applied threshold. For lower thresholds, *degree centrality* and *eigenvector centrality* exhibit slightly higher NPVR values with increased variability across regions. In contrast, for *clustering coefficient* and *betweenness centrality*, higher thresholds tend to produce substantially larger NPVR magnitudes, accompanied by pronounced regional outliers. The spatial variability is also greater for these metrics at higher

thresholds, reflecting more localized and extreme numerical variability.

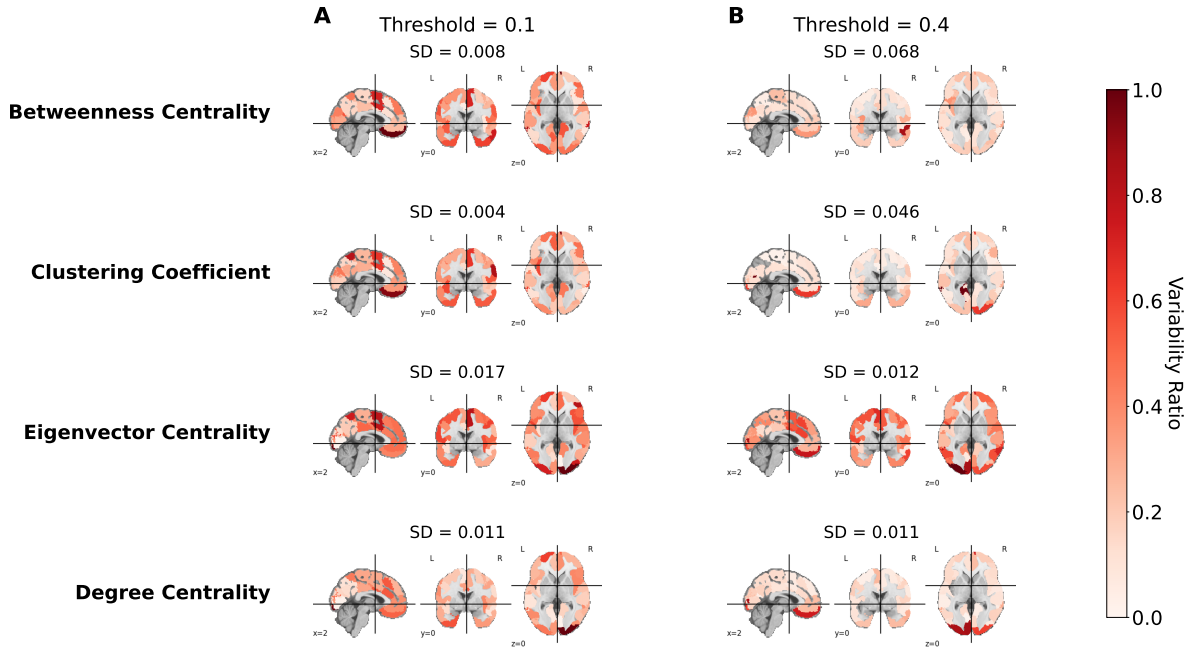


Figure 3.4: **Numerical-to-population variability ratio across brain regions.** Normalized regional numerical-to-population variability ratio (NPVR) maps for the pooled Parkinson's disease (PD) and healthy control (HC) sample at **(A)** threshold 0.1 and **(B)** threshold 0.4. Rows show betweenness centrality, clustering coefficient, eigenvector centrality, and degree centrality. Values are normalized independently within each metric and threshold, with darker red indicating a higher NPVR. SD denotes the regional standard deviation of the NPVR.

This result highlights that numerical variability is not uniformly distributed across brain regions and that certain regions or metrics are more sensitive to thresholding. Consequently, inference drawn from regional graph metrics—such as node-level comparisons or correlations with behavior—may be differentially affected by numerical variability depending on both the metric and the chosen threshold.

Regional NPVR maps were also generated separately for the PD and HC groups and are provided in Figs A and B in S1 File. The same key conclusions applied across both populations: NPVR exhibited a spatially heterogeneous distribution, and this heterogeneity varied with both the graph metric and the graph threshold.

Trade-offs of using confound removal

To assess the influence of confound regression on the numerical variability of the graph metrics, we compared the results obtained from the *with-confound* and *no-confound* pipelines. Figs. 3.5 and 3.6 show the difference between the two processing pipelines, calculated as *no-confound* minus *with-confound*. The differences are shown across thresholds for the pooled PD+HC sample and separately for the PD and HC groups. For local graph metrics, the stars represent the region-averaged NPVR differences, whereas for global graph metrics they represent the corresponding NPVR differences for each metric. Negative values indicate lower NPVR in the *no-confound* pipeline.

In Fig. 3.5, for each metric and threshold, the NPVR difference was first calculated separately for each of the 100 regions as *no-confound* minus *with-confound*. Permutation tests were then applied to these regional differences to determine whether their distribution differed significantly from zero. The stars represent the mean of the 100 regional NPVR differences and were predominantly negative, indicating lower NPVR in the *no-confound* pipeline relative to the *with-confound* pipeline. Most metric–threshold combinations showed a statistically significant negative difference, although some comparisons were not significant. Overall, the tendency toward lower NPVR in the *no-confound* pipeline was reproduced across populations, although the magnitude and, at some thresholds, the direction of the effect differed between PD and HC.

For the global graph metrics (Fig. 3.6), the pooled PD+HC, PD, and HC samples also showed predominantly negative NPVR differences. However, positive differences were observed at $T = 0.1$ for average shortest path length, whereas the differences were negative at most other thresholds. These results indicate that the effect of confound regression on numerical variability is generally consistent with the pooled analysis but can depend on population, graph metric, and network threshold.

However, this reduction in NPVR does not necessarily imply better validity: removing confound regression may complicate the interpretation of the resulting connectivity estimates and could compromise the accuracy of the derived functional connectomes.

Additional results for the combined PD+HC sample and the separate PD and HC groups, showing differences in population variability, numerical variability, and regional NPVR between the no-confound and with-confound pipelines across local graph metrics, are provided in Figs C–H in S1 File.

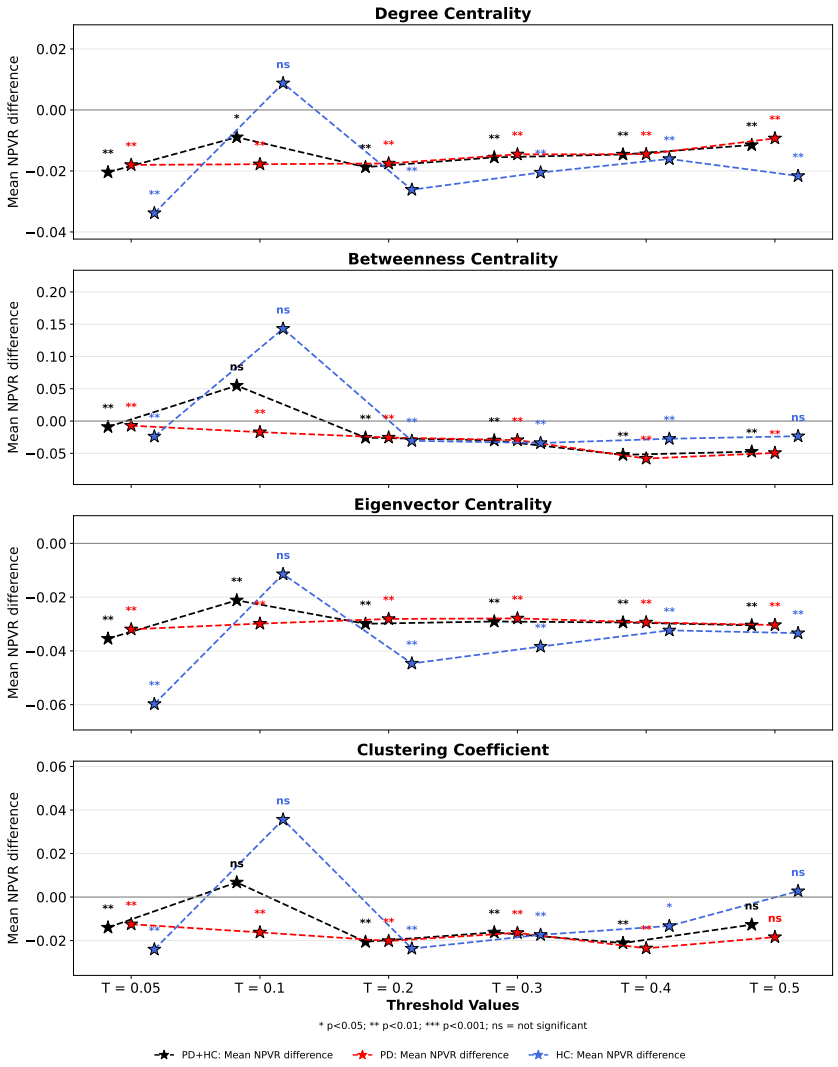


Figure 3.5: **Numerical-to-population variability ratio differences across pipelines for local graph metrics.** Region-averaged numerical-to-population variability ratio (NPVR) differences between the *no-confound* and *with-confound* pipelines, calculated as *no-confound* minus *with-confound*. Results are shown for the pooled Parkinson's disease (PD) and healthy control (HC) sample (black), PD group (red), and HC group (blue). All populations were evaluated using the same network thresholds: $T = 0.05, 0.10, 0.20, 0.30, 0.40$, and 0.50. Markers were shifted slightly along the horizontal axis to prevent overlap and improve visualization; these offsets do not represent different threshold values.

NPVR affects downstream statistical inference

Fig. 3.7 shows the estimated probability that numerical variability changes a result's significance classification as a function of the distance between the nominal p-value and the significance threshold, $p_0 - \alpha$, with $\alpha = 0.05$. Each point represents one configuration defined by a graph metric, network threshold, sample size, representative test statistic, and statistical

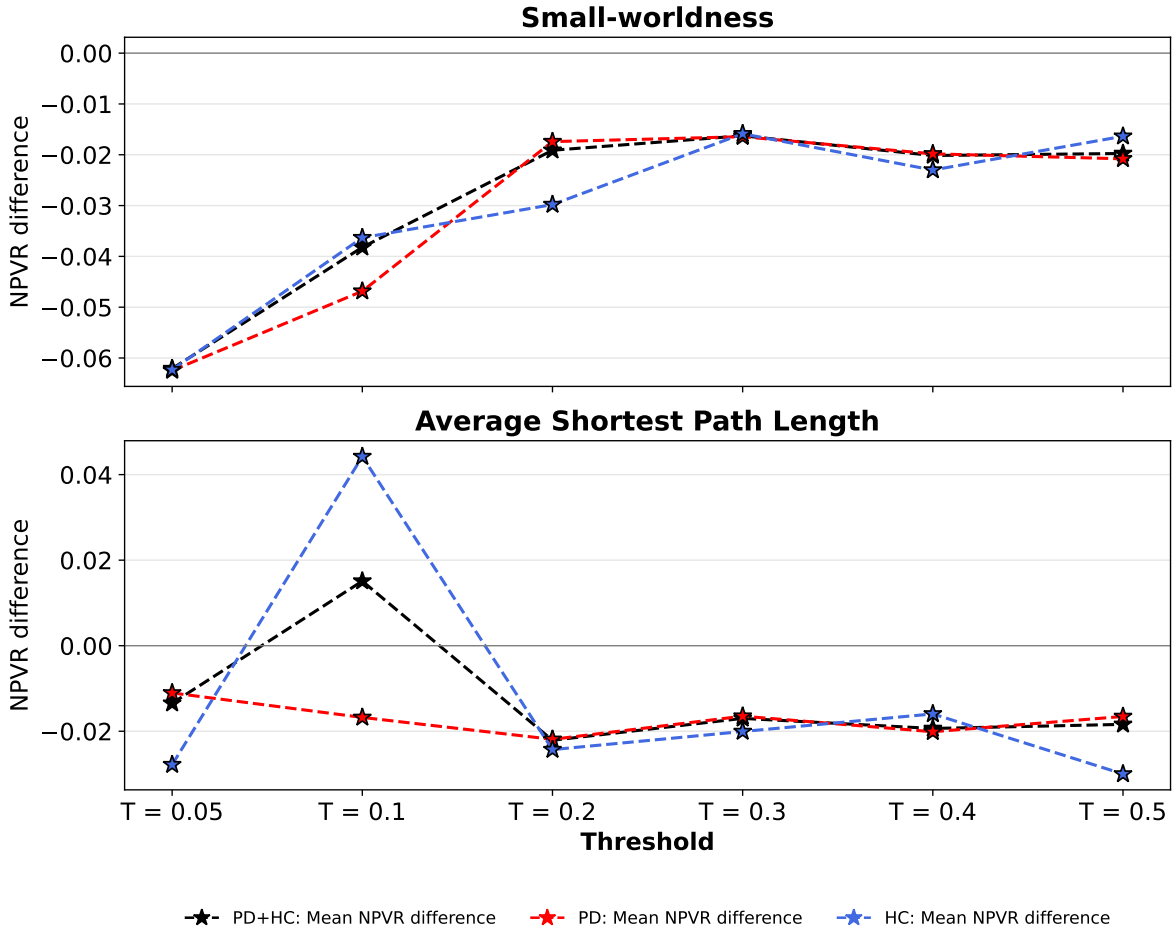


Figure 3.6: **Numerical-to-population variability ratio differences across pipelines for global graph metrics.** Numerical-to-population variability ratio (NPVR) differences between the *no-confound* and *with-confound* pipelines, calculated as *no-confound* minus *with-confound*, for small-worldness and average shortest path length across network thresholds. Results are shown for the pooled Parkinson's disease (PD) and healthy control (HC) sample (black), PD group (red), and HC group (blue). All populations were evaluated using the same network thresholds: $T = 0.05, 0.10, 0.20, 0.30, 0.40$, and 0.50. Markers were shifted slightly along the horizontal axis to prevent overlap and improve visualization; these offsets do not represent different threshold values.

test. The configurations include two-sample t-tests, partial correlations, and ANCOVA. Negative values of $p_0 - \alpha$ correspond to nominally significant results, for which a flip occurs when numerical variability moves the p-value above α. Positive values correspond to nominally non-significant results, for which a flip occurs when numerical variability moves the p-value to or below α.

In the pooled PD+HC analysis, the 972 configurations for which $p_0 - \alpha$ was between -0.04 and 0.04, corresponding to nominal p-values between 0.01 and 0.09, had a mean

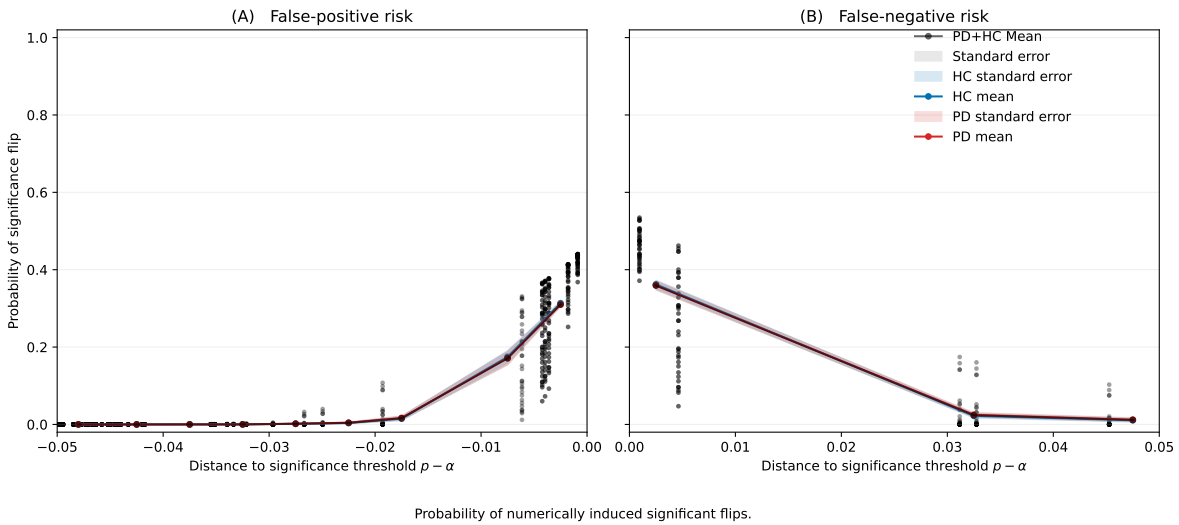


Figure 3.7: **Probability of numerically induced significance flips.** Significance-flip probability as a function of the distance between the nominal p-value and the significance threshold, $p_0 - \alpha$, with $\alpha = 0.05$. Individual points represent configurations from the pooled Parkinson's disease and healthy control (PD+HC) analysis across graph metrics, network thresholds, sample sizes, representative test-statistic values, and statistical tests. The left panel(A) shows nominally significant results that may become non-significant, whereas the right panel (B) shows nominally non-significant results that may become significant. Lines represent mean flip probabilities calculated in bins of width 0.005 for the pooled PD+HC sample (black), Parkinson's disease (PD) group (red), and healthy control (HC) group (blue). Shaded regions represent the corresponding standard errors. The false-positive and false-negative labels describe instability of the nominal significance classification rather than classical error rates based on known ground truth.

significance-flip probability of 17.7%. The flip probability was highest near $p_0 - \alpha = 0$ and decreased as the nominal p-value moved farther from the significance threshold. Thus, results close to α were the most susceptible to numerically induced changes in significance classification.

Elevated flip probabilities occurred at both small and large evaluated sample sizes when the nominal p-value remained close to α. For example, for eigenvector centrality at a network threshold of 0.05 and $t = 2$, the nominal p-value was 0.0546 and the estimated flip probability was 0.4468 at $n = 30$. At $n = 1000$, the nominal p-value was 0.0458 and the estimated flip probability was 0.3630. These results show that a large sample size does not by itself guarantee numerical stability when the nominal result lies close to the significance threshold.

The propagation formulas used to calculate σ_p are provided in Table B in S1 File. The complete configuration-level results underlying Fig. 3.7 are provided in Table C in S1 File, which reports the graph metric, network threshold, NPVR, sample size, representative test

statistic, nominal p-value, propagated σ_p, and significance-flip probability. These values can be used as practical reference points for identifying results that may be sensitive to numerical variability. They should not be interpreted as universal thresholds because numerical variability depends on the study population, acquisition protocol, preprocessing pipeline, graph-construction choices, graph metric, and computational environment. Whenever possible, NPVR should therefore be estimated for the specific pipeline and outcome under consideration.

Discussion

The reliability of quantitative network neuroscience depends not only on accurate mapping of functional connectivity but also on the stability of the derived measures. While methodological discussions typically emphasize sample size, biological variability, and effect size, measurement noise from computational sources remains largely unaddressed. We showed that numerical variability originating from floating-point arithmetic can influence functional connectomes and graph metrics. Through the Numerical-Population Variability Ratio (NPVR), we quantified this computational noise relative to inter-subject differences, framing numerical variability in statistical terms commonly used to evaluate biological effects.

Our findings reveal that numerical variability is not uniform and varies across metrics, thresholds, and spatial locations. For several metric–threshold combinations, the observed average NPVR values were on the order of 0.1 to 0.2, indicating that numerical variability accounts for a measurable fraction of the variability observed within a population. This suggests that numerical variability may influence downstream analyses, particularly when measuring small effects.

We observed similar NPVR value ranges in structural MRI [28], suggesting that this interval may be a good ballpark estimate of numerical variability in MRI analyses. This range provides a useful reference point for contextualizing results in the current literature and for approximating the variability that propagates into downstream analyses, including its impact on statistical inferences such as effect sizes, F-statistics, and other measures as proposed by [28].

The impact of numerical variability on downstream statistical inference depends on both the magnitude of the NPVR and the statistical analysis being performed. Our results show that findings near the significance threshold may be susceptible to numerically induced significance flips across sample sizes, highlighting that numerical variability is not only a technical artifact but a substantive factor that can directly influence scientific conclusions and cannot be mitigated solely by increasing sample size. NPVR, together with its propagation to

the downstream analyses, provides a practical framework for evaluating numerical variability and estimating its potential contribution to false-positive and false-negative risks.

However, the specific NPVR values reported in our study will likely vary across multiple factors. Numerical variability is inherently shaped by the chosen processing pipeline, the task or resting-state paradigm, sample characteristics, and the neurobiological properties of the studied population. Prior work has already demonstrated that certain preprocessing steps introduce substantial numerical instability: for example, [27] showed that linear registration exhibits marked numerical variability across runs, and [26] demonstrated that the FastSurfer deep-learning pipeline does not yield lower numerical variability than the traditional FreeSurfer workflow. These findings reinforce that numerical variability is not tied to a specific algorithmic class and may emerge across both conventional and machine-learning–based pipelines. Understanding how NPVR interacts with specific preprocessing steps, graph-construction choices, or distinct subject cohorts may ultimately become informative in its own right. In particular, if certain pipelines or populations consistently exhibit elevated NPVR in specific regions or metrics, this could motivate novel quality-control strategies or even the development of computational biomarkers. A systematic investigation of these interactions would therefore be an important direction for future work. Moreover, although graph measures such as small-worldness have been associated with altered network organization in PD, the present study was not designed to distinguish biological alterations from population-specific differences in NPVR values. Furthermore, we focused on cross-sectional studies, where numerical variability is evaluated relative to inter-subject differences. How longitudinal pipelines behave in fMRI analysis remains an open question and an important direction for future research. Furthermore, restricting the analysis to RL acquisitions reduced acquisition-related heterogeneity but may limit the generalizability of the reported NPVR estimates to other phase-encoding directions and acquisition protocols. Future work should evaluate whether NPVR remains consistent across acquisition configurations.

Approximately 80% of network studies apply some form of thresholding to the connectivity matrix, making it common in both structural and functional network analyses. Finding the optimal thresholding approach remains an open question [33].

Future work could explore how network sparsification and different thresholding methods influence the NPVR of graph measures. Our results suggest that intermediate thresholds can reduce NPVR for some graph metrics, although the threshold associated with minimum NPVR is metric-dependent.

Other post-processing choices also influence the distribution of numerical variability. Motion-related and physiological regressors alter the regional time series and subsequent

connectivity estimates, which may change how floating-point perturbations propagate through graph construction; characterizing this interaction mechanistically remains an important direction for future work. Omitting confound regression consistently lowers NPVR, but this does not necessarily imply more accurate estimates. Numerical stability is only one dimension of pipeline performance: a method can be highly stable yet produce inaccurate results.

Reported NPVR means can be leveraged to evaluate how processing choices interact with expected effect sizes and available sample sizes, thereby informing processing decisions and the selection of thresholds that are more likely to yield robust findings.

Similarly, factors such as atlas selection should be considered from the perspective of numerical variability. As noted in [33], more detailed atlases introduce more parallel tests and may generally show greater measurement error in observed connectivity.

Overall, our results position numerical variability as an important factor in network-neuroscience studies, particularly when investigating subtle effects or interpreting statistical results close to a decision threshold.

Conclusion

In this study, we examined the numerical variability of fMRI-derived graph metrics using data from the Parkinson's Progression Markers Initiative (PPMI) cohort. We quantified numerical variability relative to inter-subject differences using the Numerical–Population Variability Ratio (NPVR). Across graph metrics, NPVR values typically ranged between 0.02 to 0.19, indicating that numerical variability accounts for a non-negligible fraction of population-level variability.

The magnitude of NPVR varied across graph metrics, brain regions, network thresholds, populations, and confound-regression strategies.

Propagating the observed NPVR values to commonly used statistical tests demonstrated that numerical variability can affect downstream significance classifications. In the pooled PD+HC analysis, among the 972 evaluated configurations with nominal p-values between 0.01 and 0.09, the mean probability of a numerically induced significance flip was 17.7%. This probability was highest for results closest to the significance threshold and remained elevated in some configurations at both small and large sample sizes. Numerical fragility therefore depends jointly on NPVR, the statistical test and test statistic, sample size, and the distance between the nominal p-value and the significance threshold; increasing sample size alone does not necessarily eliminate it.

These findings motivate three complementary strategies for mitigating the impact of numerical variability. First, statistical inference can be strengthened relative to numerical noise through robust study design, high-quality data acquisition, appropriate quality control, and, where relevant, adequate sample sizes. Although increasing sample size reduces numerical uncertainty for some estimators, such as Cohen's d, it should not be treated as the sole mitigation strategy. Second, numerical variability should be systematically evaluated and reported. Repeated perturbation-based analyses can be used to estimate numerical variability, calculate NPVR, and propagate it to quantities such as σ_p and P_{flip}, allowing researchers to assess the numerical sensitivity of their conclusions. Third, numerical variability should be reduced at its source through the development and validation of more stable algorithms, software libraries, processing pipelines, and computational methods.

Together, these strategies can improve the transparency and numerical reliability of functional network analyses. Explicitly evaluating numerical variability alongside biological and sampling variability will help ensure that conclusions derived from fMRI graph measures remain robust to computational perturbations.

Chapter 4

Future Work: Predicting Numerical Variability

4.1 Motivation and main question

The first phase of this research demonstrated that numerical variability (NV) in functional MRI graph measures is measurable and can influence statistical conclusions. These findings establish the importance of evaluating numerical stability, particularly when the observed effects are small or close to a decision threshold. However, direct assessment with Monte Carlo Arithmetic (MCA) requires repeated executions of the processing workflow and is therefore computationally expensive. In addition, most researchers do not routinely have access to an MCA-instrumented environment. Consequently, numerical stability is rarely evaluated as part of a conventional neuroimaging analysis.

The next phase of this research is motivated by a practical question:

> *Can the numerical variability of graph measures be predicted from the functional connectome produced by a single standard execution?*

Here, a standard or *IEEE* execution refers to ordinary processing without MCA perturbations. If the resulting functional connectivity (FC) matrix contains information about its own numerical sensitivity, it may be possible to estimate NV without repeating the complete workflow. Such an estimate would not replace MCA as the reference method for characterizing numerical uncertainty, but it could provide an accessible screening tool for identifying potentially unstable results.

4.2 Why prediction may be possible

Preliminary exploratory analyses investigated how MCA perturbations in FC propagate into graph-metric NV. The findings suggest that this propagation is structured rather than uniform. In particular, connections whose correlation values are close to the graph-construction threshold are more likely to switch between being included and excluded. These edge changes can alter network topology and amplify small numerical perturbations into larger differences in graph measures.

Threshold proximity alone does not fully determine this effect. The spatial and topological locations of the affected edges are also important: a perturbation involving a highly connected or otherwise influential part of the network may have a different consequence from an equally sized perturbation elsewhere. The analysis further indicated that binary graph variability arises through changes in topology, whereas weighted graph variability can reflect both changing topology and changing connection weights. Together, these findings suggest that the standard connectome contains potentially informative features—including connection magnitude, distance from the threshold, and the organization of vulnerable edges—that may indicate future numerical sensitivity. This provides the main rationale for prediction.

4.3 Preliminary prediction results

Initial experiments used a simple BrainNetCNN-style model to predict graph-measure NV from the IEEE connectivity matrix. This model was selected because its edge-to-edge and edge-to-node operations are designed to retain relationships among connections and brain regions rather than treating connectivity values as unrelated features. A logarithmic transformation of NV was used to reduce the influence of its strongly skewed distribution.

The results provide preliminary evidence that prediction may be feasible, although performance is not consistent across all graph measures or thresholds. Some outcomes, including degree and eigenvector centrality in lower-threshold graphs and betweenness or clustering in other threshold settings, showed more promising prediction than others.

For several graph measures, the Spearman correlations between predicted and observed NV were informative even when prediction of the exact NV magnitude remained limited. This indicates that the model may already contain useful information for ordering participants from less to more numerically sensitive. Such ordering is relevant in practice because it could help identify participants for whom a direct MCA analysis should be prioritized.

These findings remain preliminary and do not yet establish a generally accurate predictor.

Their importance is instead that they support the central premise of this phase: the IEEE connectome appears to contain information about numerical fragility, but the usefulness of that information differs across graph measures and graph-construction choices.

4.4 Challenges and proposed direction

Predicting NV presents several conceptual challenges. The response to MCA perturbations is itself random, and NV estimated from a finite number of realizations is therefore uncertain. This uncertainty can limit the apparent accuracy of a model even when a relationship with the underlying numerical sensitivity exists. In addition, the IEEE connectome may not contain all relevant information about how a processing workflow responds to perturbation. Numerical sensitivity may also depend on intermediate computations or computational conditions that are not represented in the final FC matrix.

The next stage will investigate whether prediction can be improved. However, exact numerical prediction is not the only potentially useful outcome. If the uncertainty of the target or missing sensitivity information places a practical limit on magnitude prediction, the work will evaluate whether participants can instead be ranked reliably by NV or classified as having comparatively high numerical sensitivity. Identifying highly sensitive participants would still provide a practical basis for directing computationally expensive MCA analyses toward the cases in which they are most needed.

4.5 Expected contribution

This phase will determine whether information available from a single standard connectome can be used to anticipate the numerical fragility of derived graph measures. It will connect the mechanistic evidence on threshold crossing and network topology with a practical prediction problem. A successful approach could make numerical-stability assessment more accessible by providing either an estimate of NV or a reliable indication of which participants are most likely to be sensitive. If prediction remains limited, the results will still clarify which aspects of numerical sensitivity are absent from the standard connectome and where direct perturbation-based evaluation remains necessary.

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